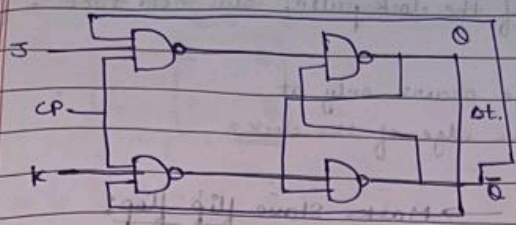
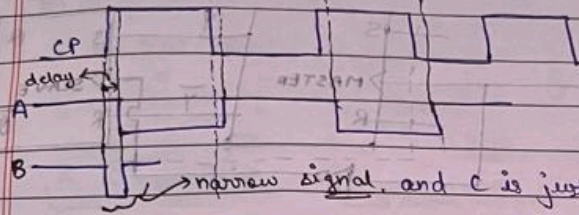
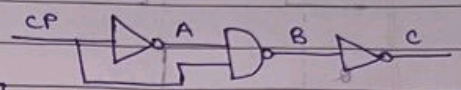


- 3 ways to modify:-
- ① Ensure $\Delta w < \Delta t$ (using narrow clock) not in our hand.
 - ② Edge Triggered flip-flop (Circuit is triggered either in +ve or -ve direction)
 - ③ Master-Slave flip-flop (when 1 is active, other is inactive).
- (2 flip flops)

NAND Version:

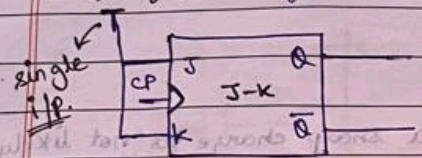


Generating narrow clock signal:



You can use odd no of inverters for more delay and same o/p.

② T-flip-flop: Modification of J-K.



Q(t)	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

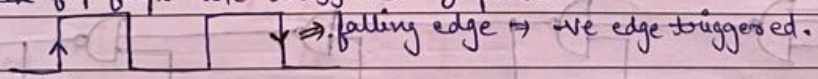
When $T=0 \Rightarrow$ no change.
When $T=1 \Rightarrow$ toggling occurs.
and $Q(t+1)$ is complement of $Q(t)$.

$Q(t+1) = \overline{T}Q(t) + T\overline{Q}(t)$

Toggle-flip-flop.

→ The state of a flip-flop is changed by momentary change in the i/p signal. This momentary change is called a trigger.

• ~~clock~~ flip flops are triggered by pulse.



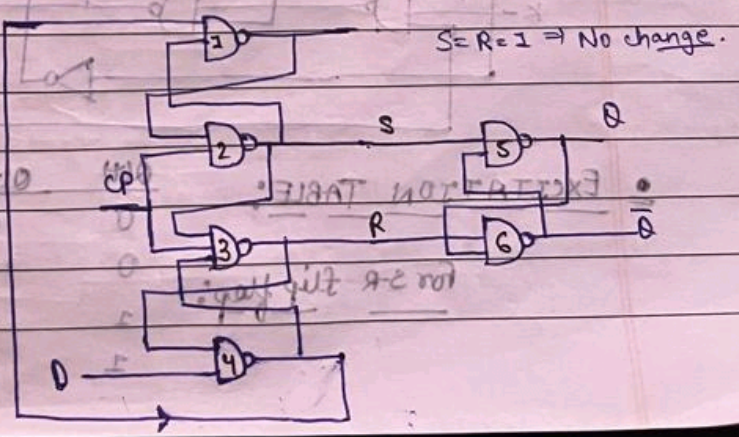
if triggered at rising edge of clock $\Rightarrow$ +ve edge triggered.

Edge Trigger D-flip-flop:

(3 latches). Logic Diagram:

+ve edge triggered.

S	R	Q	Q-bar
1	1	no change	
0	1	1	0
1	0	0	1



average by groups is

19/09/24

The Excitation Table:

P.S. (y)	X ₁	X ₂	N.S.	D	Z = output
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	1	1

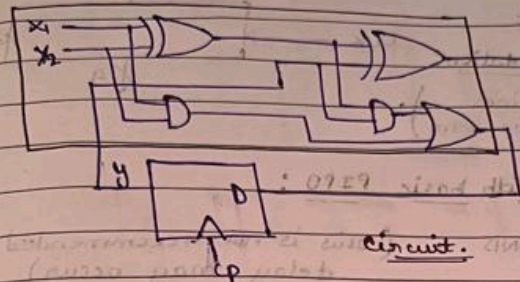
Expression:

$$Z = y \oplus X_1 \oplus X_2$$

$$D = X_1 X_2 + X_1 Y + X_2 Y$$

(same as carry of full adder.)

Full Adder



Circuit.

Design a sequence detector that detects '0120' and also the overlapping occurrences.

overlapping occurrences.

P.S.	N.S.	O/P	N.S.	O/P
(00) A	B	0	A	0
(01) B	B	0	C	0
(10) C	B	0	D	0
(11) D	B	1	A	0

4 states $\Rightarrow$ 2 flip flops. (2 T we can use)

The Excitation Table:-

X (ip)	Y ₁	Y ₂	Y ₁	Y ₂	T ₁	T ₂	Z
0	0	0	0	1	0	1	0
0	0	1	0	1	0	0	0
0	1	0	0	1	1	1	0
0	1	1	0	1	1	0	1
1	0	0	0	0	0	0	0
1	0	1	1	0	1	1	0
1	1	0	1	1	0	1	0
1	1	1	0	0	1	1	0

$$\Rightarrow Z = \bar{X} Y_1 Y_2$$

$$T_1 = \bar{X} Y_1 \bar{Y}_2 + \bar{X} Y_1 Y_2 + X \bar{Y}_1 Y_2 + X Y_1 Y_2$$

$$\Rightarrow T_1 = \bar{X} Y_1 + X Y_2$$

$$T_2 = \bar{X} \bar{Y}_1 \bar{Y}_2 + X \bar{Y}_1 Y_2 + X Y_1 Y_2 + X Y_1 \bar{Y}_2 + X Y_1 Y_2$$

15/10

Design of Registers:

A register consists of group of flip-flop with common clock signal, used to store binary information. They are of various types:

- Parallel-In-Parallel Out (PIPO)
- Serial-in-Serial-Out (SISO)
- Parallel-In-Serial Out (PISO)
- Serial-in-Parallel-Out (SIPO)

$\rightarrow$ If a register supports serial-in or serial out or both modes, then register is called the shift register.

09/09/23

Step ⑤ until no more 'x' can be added. (5) for a square state.

doesn't contain the 'x' mark, the states s_i and s_j are equivalent.

→ 1st iteration
→ 2nd iteration
→ if o/p is not same
(self-implied, so cancel)

(a-a), (e-e)
not considered as both are same state.

PS.	N.S.	X=0	X=1	P.O.
a	a	c	0	0
b	f	h	0	0
c	e	a	1	0
d	a	e	1	1
e	c	a	1	1
f	f	b	0	0
g	b	h	0	1
h	c	g	1	1

check 2nd row.

(No need new)

In exam, draw a table for each iteration.

After 2nd iteration, we observe no more 'x' can be added.

$\Rightarrow a \equiv d, c \equiv e$

Merging equivalent states now: see in above table.

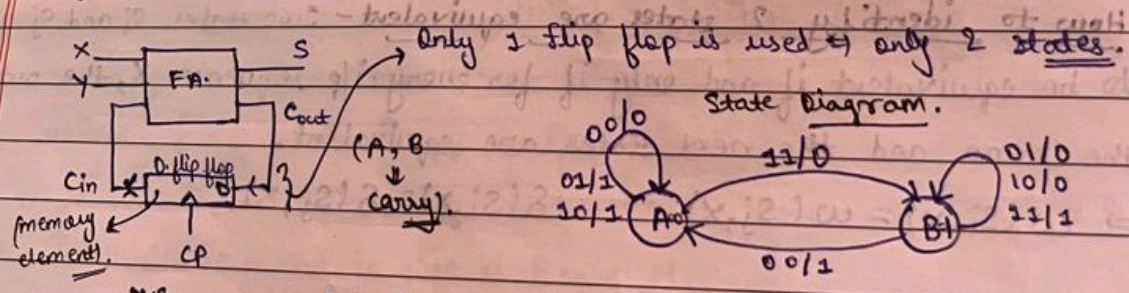
Reduced table:

PS.	N.S.	X=0	X=1	P.O.
a	a	c	0	0
b	f	h	0	0
c	c	a	1	0
f	f	b	1	1
g	b	h	0	0
h	c	g	1	1

State Assignment:

a = 000
b = 001
c = 010
f = 011
g = 100
h = 101

Design a serial adder:



State table:

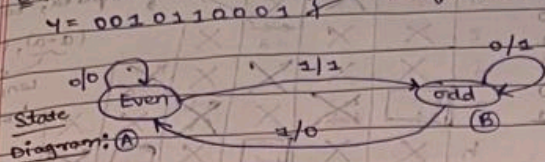
PS.	00	01	10	11	Output
0	0	0	0	1	0 1 1 0
1	0	1	1	1	1 0 0 1

D flip flop is used as memory element.

- ② Design a serial parity detector. A continuous stream of bits is fed to a ckt. in synchronism with clock the ckt will generate a bit stream as o/p, where a 0 will indicate even no. of 1's seen so far and a 1 will indicate odd no. of 1's seen so far.

Ex $X = 0011101001$
 $Y = 0010110001$

No. of states = 2 (2 odd & 2 even).

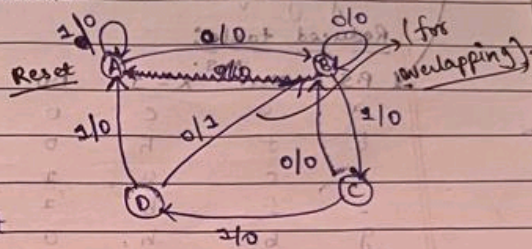


State Table:

PS	X=0		X=1	
	N.S.	O/P	N.S.	O/P
Even A	A	0	B	1
Odd B	B	1	A	0

- ③ Design a sequence detector. A circuit accepts a serial bit stream X as i/p and produces a serial bit stream Z as o/p. Whenever the pattern 0110 appears in the i/p stream, its o/p Z=1 and at all other times Z=0. Overlapping occurrences of the patterns are also detected.

$X = 00111001001101100$
 $Z = 0000100000010010$



State Table Minimization:

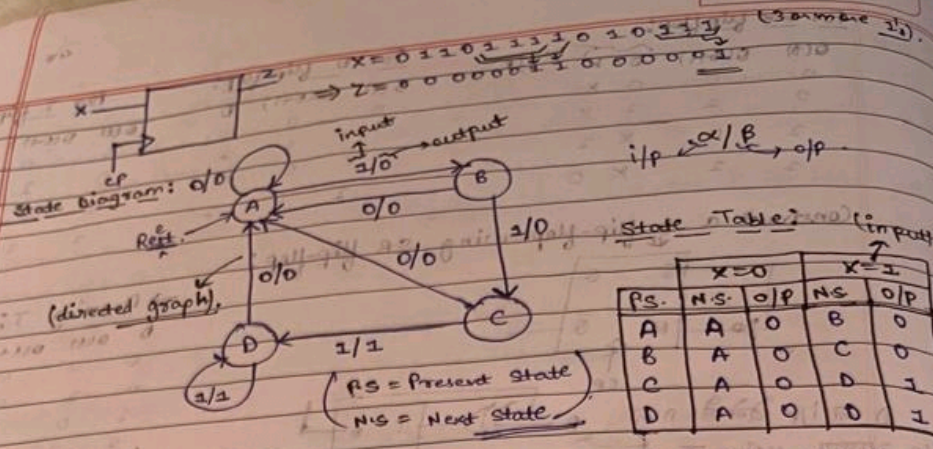
Basic Concept - Identify equivalent states and merge them into a single state. This leads to more compact representation.

How to identify 2 states are equivalent - Two states S_i and S_j are said to be equivalent if and only if for every i/p sequence X, the outputs are the same and the next states are equivalent.

$$\Rightarrow w(S_i, X) = w(S_j, X) \quad \delta(S_i, X) \equiv \delta(S_j, X)$$

→ Steps for minimization of state table using implication table:

- Construct an implication table that contains a square for every pair of states.
- Compare each pair of row in the state table, if the o/p associated with the states S_i, S_j are different, place a 'X' in corresponding square $S_i - S_j$ to indicate S_i and S_j are not equivalent.
- If the outputs are same, place the implied pair in the square $S_i - S_j$.
- Go through the table square by square. If square $S_i - S_j$ contains the implied pair 'm-n' and the square 'm-n' contains a 'X' mark, then S_i, S_j are not equivalent, and place a 'X' in square $S_i - S_j$.
- If any 'X' is added in step (3), repeat

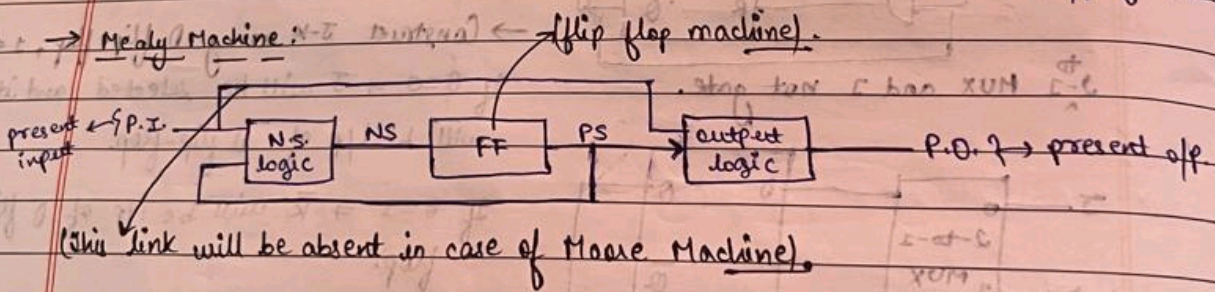


→ Formal Definition of FSM: A deterministic FSM can be mathematically defined as a 6 tuple (S, Y, S_0, δ, w) .

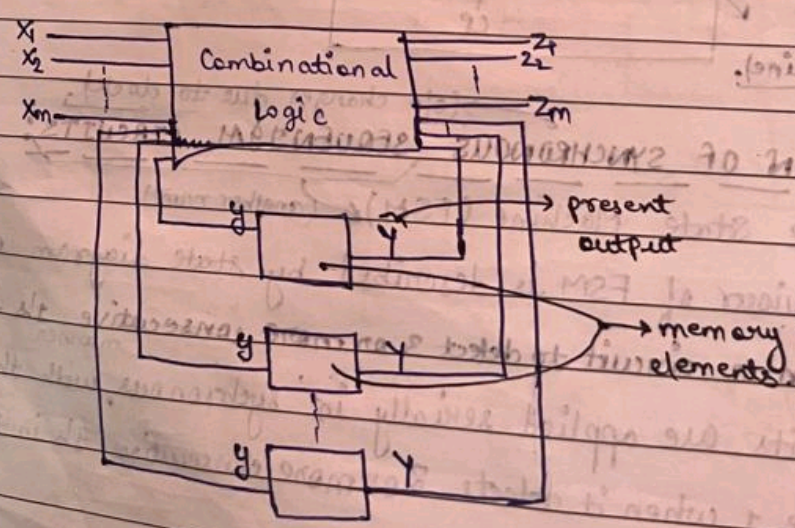
- $S \Rightarrow$ set of input combination (alphabet Γ / $Y \Rightarrow$ set of output combination)
- $S_0 \Rightarrow$ It is the initial state. ($S_0 \in S$)
- $\delta \Rightarrow$ is the state transition function $\rightarrow$ gives the next state.
- $w \Rightarrow$ output function.

→ 2 types of FSM:

- Mealy: output depends on both the present state as well as i/p.
- Moore: o/p depends only on i/p of machine.



→ Block diagram of general FSM:



for S-R flip-flop:

S	R	Q(t)	Q(t+1)
0	0	0	0
0	1	1	0
1	0	0	1
1	1	X	X

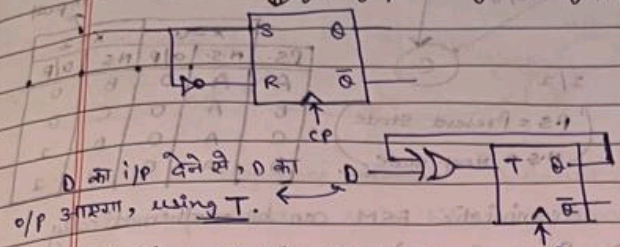
for D flip-flop:

D	Q(t)	Q(t+1)
0	0	0
0	1	0
1	0	1
1	1	1

for T flip-flop:

T	Q(t)	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

Construct a D flip-flop using S-R flip-flop:



→ D using T:

D	Q(t)	Q(t+1)	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

characteristic table

$$T = D \oplus Q(t)$$

excitation table

Using S-R flip-flop:

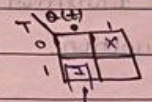
S	R	Q(t)	Q(t+1)	T
0	X	0	0	0
1	0	0	1	1
0	1	1	0	1
X	0	1	1	0

characteristic table

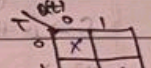
$$S = T\bar{Q}(t), R = TQ(t)$$

excitation

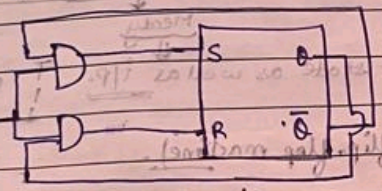
इस order में बनाना, जो कौन सा दो i/p देता है।



$$S = T\bar{Q}(t)$$

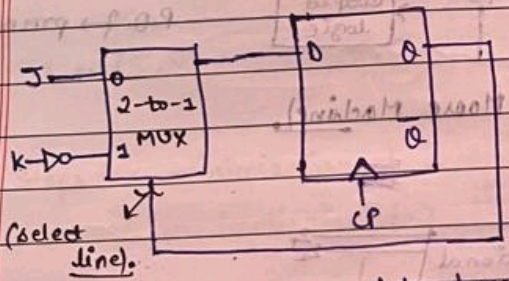


$$R = TQ(t)$$



Block diagram.

2-to-1 MUX and 1 not gate.



→ Construct J-K using D flip-flop, 1 to 0 if Q=0 ⇒ J will be selected and it will be i/p of D flip-flop.

if Q=1 ⇒ K will be i/p of D flip-flop.

DESIGN OF SYNCHRONOUS SEQUENTIAL CIRCUITS:

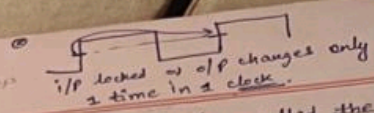
Finite State Machine (FSM) (another name)

→ Behaviour of FSM is described by state diagram or state table.

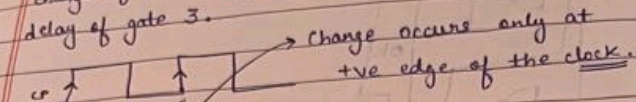
→ Design a circuit to detect 3 or more consecutive 1's in a serial bit stream:

The bits are applied serially in synchronous manner with the clock. The o/p will become 1 when it detects 3 or more consecutive 1's in the stream else 0.

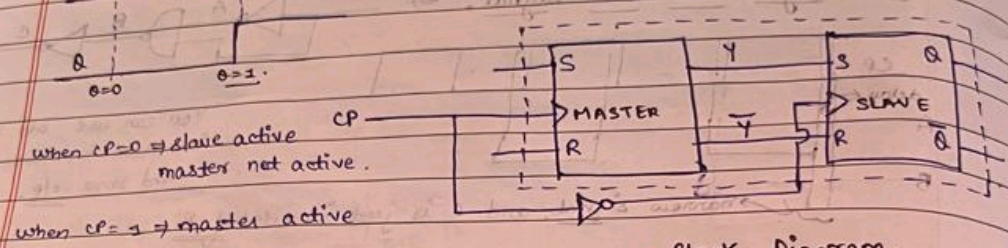
Combinational dkt.
Behaviour by
truth table.



- There is a definite time called the setup time in which the D i/p must be maintained at a constant value (level) prior to the application of CP. That setup time is equal to the propagation delay through gates 4 and 5.
- There is a definite time called the hold time that the D i/p must not change after the application of the clock pulse. This hold time $\equiv$ propagation delay of gate 3.



Master Slave flip-flop:



Block Diagram.

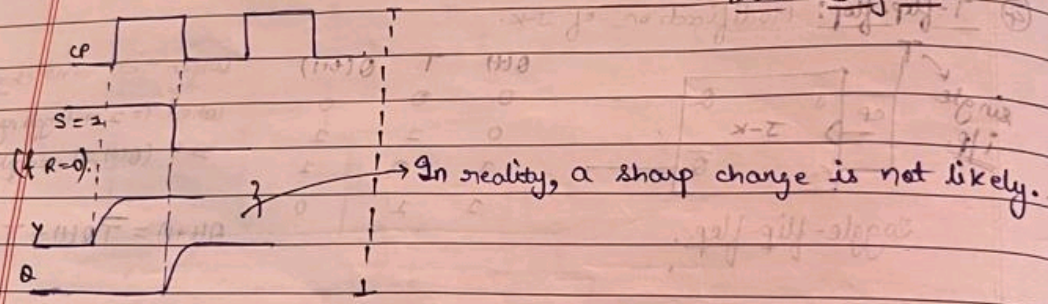
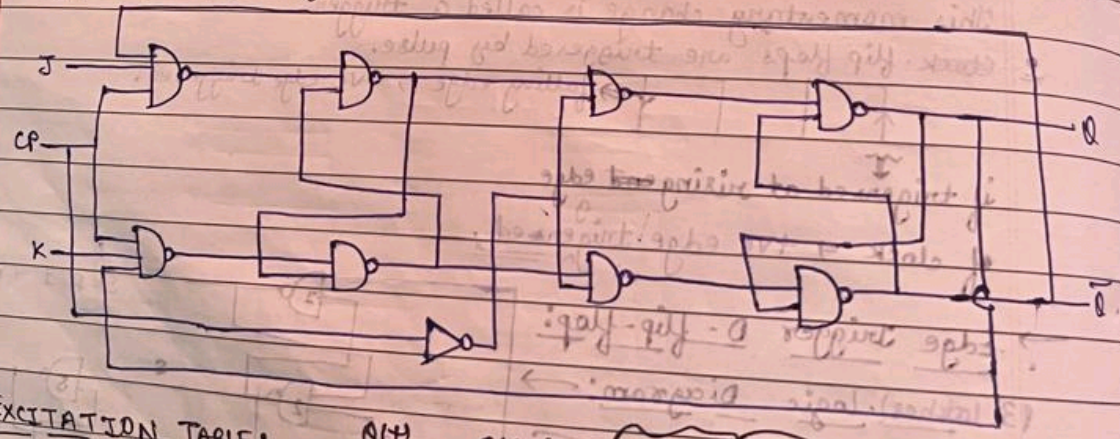


Diagram for J-K flip flop:



EXCITATION TABLE:

for S-R flip flop:

Q(t)	Q(t+1)	S	R	O/P.
0	0	0	X	
0	1	1	0	
1	0	0	1	
1	1	X	0	